

INDEX

HelixCode User Manual - Quick Index

Version: 2.1 | Date: May 8, 2026 | Total: 7,549 lines



Main Documentation

[README.md](#) - 3,027 lines

Comprehensive user manual covering all HelixCode features

Quick Navigation: - [Introduction](#) - What is HelixCode, architecture, use cases - [Installation](#) - Quick start, production, Docker, Kubernetes - [First Project](#) - Create your first application - [Configuration](#) - YAML config, environment variables - [LLM Providers](#) - All 14+ providers documented - [Anthropic Claude](#) - Extended thinking, caching - [Google Gemini](#) - 2M token context - [OpenAI](#) - GPT-4.1, O1 reasoning - [AWS Bedrock](#) - Enterprise multi-model - [Azure OpenAI](#) - Entra ID, compliance - [VertexAI](#) - GCP unified platform - [Groq](#) - Ultra-fast 500+ tok/s - [Mistral](#) - EU data residency - [XAI Grok](#) - Free tier - [OpenRouter](#) - 100+ models - [GitHub Copilot](#) - Free with subscription - [Qwen](#) - 2K free requests/day - [Ollama](#) - 100% local - [Llama.cpp](#) - Direct inference - [Core Tools](#) - [File System](#) - [Shell](#) - [Browser](#) - [Web](#) - [Voice](#) - [Codebase Mapping](#) - [Advanced Workflows](#) - [Plan Mode](#) - [Multi-File Editing](#) - [Git Auto-Commit](#) - [Context Compression](#) - [Tool Confirmation](#) - [Snapshots](#) - [Autonomy Modes](#) - [Vision Auto-Switch](#) - [Distributed Computing](#) - [Memory System](#) - [Sandbox Configuration](#) - [Search Provider](#) - [Security](#) - [Troubleshooting](#) - [Common Issues](#) - [SSH Worker Issues](#) - [LLM Connection Issues](#) - [Rate Limit Handling](#) - [CLI Reference](#) - [API Reference](#) - [FAQ](#)



Tutorials (10 Comprehensive Guides) - 2,514 lines

Beginner Level

[Tutorial 1: Building a Web App from Scratch](#) - 1,245 lines



45-60 minutes - Complete REST API with authentication - PostgreSQL database design - Three-layer architecture - Docker deployment - **Output:** Working Task Management API

[Tutorial 3: Using Multiple AI Providers](#) - 110 lines



20 minutes - Provider selection strategy - Cost optimization - Automatic fallback - **Outcome:** 70% cost reduction

[Tutorial 5: Voice-to-Code Workflow](#) - 118 lines



15 minutes - Hands-free coding - Voice commands - Whisper transcription - **Use Case:** Accessibility, multitasking

Intermediate Level

[Tutorial 2: Refactoring a Large Codebase](#) - 133 lines

 **30-45 minutes** - Codebase mapping (247 files) - Multi-file atomic edits - Checkpoint snapshots - **Result:** 95% test coverage in 30 min

[Tutorial 4: Browser Automation for Testing](#) - 149 lines

 **30 minutes** - E2E test automation - Visual regression testing - Web scraping - **Technology:** Chrome/Chromium

[Tutorial 6: Multi-File Atomic Edits](#) - 134 lines

 **20 minutes** - Transaction-based editing - Atomic commits with rollback - Cross-file refactoring - **Safety:** All-or-nothing

Advanced Level

[Tutorial 7: Distributed Development with Workers](#) - 165 lines

 **30 minutes** - SSH worker setup - Auto-installation - Task distribution - **Performance:** 3x speedup

[Tutorial 8: Using Plan Mode for Complex Projects](#) - 301 lines

 **45 minutes** - Two-phase planning - Option generation - Step-by-step execution - **Example:** Real-time chat app (15K LOC in 9 hours)

Advanced Level (continued)

[Tutorial 9: Creating a Custom LLM Provider](#) - 79 lines

 **30 minutes** - Implement the `Provider` interface - Provider scaffold (Name, Generate, GenerateStream) - Configuration & registration - **Skill:** Extending HelixCode

[Tutorial 10: Adding a New Tool via MCP SDK](#) - 80 lines

 **30 minutes** - Define tool schema with JSON-RPC - Implement `Execute()` business logic - Register in MCP registry - **Skill:** MCP server extension

Configuration Examples - 740 lines

[basic config.yaml](#) - 57 lines

Purpose: Development and small projects - Local Ollama provider - Minimal dependencies - Debug logging - **Best For:** Learning, prototyping

[provider_configs.yaml](#) - 174 lines

Purpose: Complete reference for all AI providers - 14+ provider configurations - API key setup - Advanced features - **Best For:** Multi-provider setups

[enterprise_setup.yaml](#) - 248 lines

Purpose: Production enterprise deployment - High availability - Security (SSO, encryption, audit) - Monitoring (Prometheus, Grafana) - Auto-scaling - Compliance (GDPR, SOX, HIPAA) - **Best For:** Large organizations

[multi_worker_setup.yaml](#) - 261 lines

Purpose: Distributed worker orchestration - 3 worker pools (build, GPU, test) - 10+ workers (x86, ARM, macOS) - Auto-scaling with cloud - Cost optimization - **Best For:** Large-scale parallel development

 Coverage Matrix

Providers Documented (14+)

Provider	Free Tier	Context	Speed	Best For
Anthropic Claude ★	✗	200K	Medium	Complex reasoning
Google Gemini ★	✗	2M	Fast	Large codebases
OpenAI	✗	128K-1M	Medium	General purpose
AWS Bedrock	✗	Varies	Medium	Enterprise
Azure OpenAI	✗	128K	Medium	Microsoft stack
VertexAI	✗	2M	Fast	GCP integration
Groq	✗	32K	Ultra	Fast iterations
Mistral	✗	128K	Fast	EU compliance
XAI Grok	✓	128K	Medium	Experimentation
OpenRouter	✓	Varies	Varies	Model variety
GitHub Copilot	✓*	128K	Fast	GitHub users
Qwen	✓**	32K	Fast	Chinese language
Ollama	✓	Varies	Medium	Privacy
Llama.cpp	✓	Varies	Fast	Performance

*Free with GitHub subscription **2,000 free requests/day

Tools Covered (6)

- ✓ File System (read, write, edit, search)
- ✓ Shell Execution (sandboxed)
- ✓ Browser Automation

-  Web Tools (search, fetch)
-  Voice-to-Code
-  Codebase Mapping (30+ languages)

Workflows Covered (8)

-  Plan Mode
-  Multi-File Editing
-  Git Auto-Commit
-  Context Compression
-  Tool Confirmation
-  Checkpoint Snapshots
-  Autonomy Modes (5 levels)
-  Vision Auto-Switch

Quick Start Paths

Path 1: Get Started in 5 Minutes

1. Read [README: Introduction](#)
2. Follow [README: Quick Start Installation](#)
3. Complete [Tutorial 3: Multiple AI Providers](#)

Path 2: Build Your First App (1 Hour)

1. Setup [basic_config.yaml](#)
2. Follow [Tutorial 1: Building a Web App](#)
3. Deploy with Docker

Path 3: Master Advanced Features (3 Hours)

1. Complete [Tutorial 2: Refactoring](#)
2. Learn [Tutorial 6: Multi-File Edits](#)
3. Master [Tutorial 8: Plan Mode](#)

Path 4: Enterprise Deployment (1 Day)

1. Review [README: Security Best Practices](#)
2. Configure [enterprise_setup.yaml](#)
3. Setup [Tutorial 7: Distributed Workers](#)
4. Deploy to Kubernetes

Documentation Structure

```
docs/user_manual/  
├── README.md           # 3,027 lines - Comprehensive manual  
├── SUMMARY.md          # 632 lines - Creation summary  
└── INDEX.md            # This file - Quick navigation
```

— tutorials/	# 2,514 lines - 10 step-by-step guides
— Tutorial_1_Building_Web_App.md	(1,245 lines)
— Tutorial_2_Refactoring_Large_Codebase.md	(133 lines)
— Tutorial_3_Multiple_AI_Providers.md	(110 lines)
— Tutorial_4_Browser_Automation.md	(149 lines)
— Tutorial_5_Voice_to_Code.md	(118 lines)
— Tutorial_6_Multi_File_Atomic_Edits.md	(134 lines)
— Tutorial_7_Distributed_Development.md	(165 lines)
— Tutorial_8_Using_Plan_Mode.md	(301 lines)
— Tutorial_9_Custom_Provider.md	(79 lines)
— Tutorial_10_Adding_a_Tool.md	(80 lines)
— examples/	# 740 lines - 4 production configs
— basic_config.yaml	(57 lines)
— provider_configs.yaml	(174 lines)
— enterprise_setup.yaml	(248 lines)
— multi_worker_setup.yaml	(261 lines)
— chapters/	# Reserved for future expansion
— images/	# Reserved for diagrams

Total: 7,549 lines of comprehensive documentation

Use Case Index

By Role

Software Developer

- [Tutorial 1: Building Web App](#)
- [Tutorial 6: Multi-File Edits](#)
- [README: Core Tools](#)

DevOps Engineer

- [Tutorial 7: Distributed Workers](#)
- [enterprise_setup.yaml](#)
- [multi_worker_setup.yaml](#)

QA/Test Engineer

- [Tutorial 4: Browser Automation](#)
- [README: Browser Automation](#)

Tech Lead/Architect

- [Tutorial 8: Plan Mode](#)
- [Tutorial 2: Refactoring](#)
- [README: Architecture Overview](#)

Accessibility Specialist

- [Tutorial 5: Voice-to-Code](#)
- [README: Voice-to-Code](#)

Enterprise Admin

- [enterprise_setup.yaml](#)
- [README: Security Best Practices](#)
- [README: Distributed Computing](#)

By Task

Code Generation

- [Tutorial 1: Building Web App](#)
- [README: LLM Providers](#)

Refactoring

- [Tutorial 2: Refactoring](#)
- [Tutorial 6: Multi-File Edits](#)

Testing

- [Tutorial 4: Browser Automation](#)
- [README: Browser Automation](#)

Deployment

- [enterprise_setup.yaml](#)
- [README: Docker Deployment](#)
- [README: Kubernetes Deployment](#)

Cost Optimization

- [Tutorial 3: Multiple Providers](#)
- [README: Cost Comparison](#)

Support Resources

Documentation

- **Main Manual:** [README.md](#)
- **Creation Summary:** [SUMMARY.md](#)
- **This Index:** [INDEX.md](#)

External Resources

- **GitHub:** <https://github.com/your-org/helixcode>
- **Documentation:** <https://docs.helixcode.dev>
- **Community:** <https://community.helixcode.dev>
- **Issues:** <https://github.com/your-org/helixcode/issues>

Getting Help

1. Check [FAQ](#) in main README
2. Search [Troubleshooting](#) section
3. Review relevant tutorial
4. Check example configurations
5. Post in community forum
6. Create GitHub issue

Version History

Version 2.0 (November 6, 2025)

-  Complete rewrite with 6,754 lines
-  14+ AI providers documented
-  8 comprehensive tutorials
-  4 production-ready configurations
-  All features covered

Version 2.1 (May 8, 2026)

-  Expanded to 7,549 lines
-  10 tutorials (added Custom Provider, MCP Tool)
-  Memory System documentation
-  Sandbox & Search Provider configuration
-  Enhanced troubleshooting coverage

Planned Updates

- Video tutorials
- Architecture diagrams
- Interactive examples
- Case studies
- Performance benchmarks



Quick Reference

Most Common Commands

Setup

```
helixcode llm provider set anthropic
helixcode llm provider set gemini --model gemini-2.5-flash
```

Development

```
helixcode plan "Create REST API"
helixcode generate "Add user authentication"
helixcode test --coverage
helixcode commit --auto
```

Advanced

```
helixcode snapshot create "Before refactoring"
helixcode edit transaction start
helixcode worker add --host worker1.example.com
```

Troubleshooting

```
helixcode health
helixcode llm providers
helixcode worker list
```

Most Useful Configurations

- **Development:** [basic_config.yaml](#)
- **All Providers:** [provider_configs.yaml](#)
- **Production:** [enterprise_setup.yaml](#)
- **Distributed:** [multi_worker_setup.yaml](#)

Last Updated: May 8, 2026 Documentation Version: 2.1 Total Lines: 7,549 Status: Complete 

Sources verified 2026-05-29: project go.mod + CLAUDE.md §3.1 (navigation index; no third-party operator instructions)

Reviewed against Constitution §11.4.99. This is a navigation/quick-index for the User Manual — it is a table of contents into README.md, the tutorials, and the example configs; it carries **no standalone third-party service setup steps** to cross-reference against a vendor doc. Notes:

- **Provider / model references** in the coverage tables and the `helixcode llm provider set ...` examples (e.g. `gemini-2.5-flash`, context-window figures, free-tier flags) are **LLMsVerifier-runtime-sourced per CONST-036** — illustrative only and may be stale; the authoritative live list is `helixcode llm models list`. Not re-pinned here (LLMsVerifier is the single source of truth).
- **External links** in the Support Resources section (`github.com/your-org/helixcode`, `docs.helixcode.dev`, `community.helixcode.dev`) are **placeholder URLs**, not live HelixDevelopment endpoints. Negative finding: left as-is (out of scope for this §11.4.99 batch; flagged for a future docs pass).
- **Version authority:** `go.mod` + CLAUDE.md §3.1 (Go 1.26 / go1.26.3 latest, PostgreSQL 15+, Redis 7+ — all confirmed current this run). No version pins in this file required a vendor fetch.